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Feasibility Study of Grid Connected PV-Biomass Integrated Energy System in Egypt

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Abstract: The aim of this paper is to present a feasibility study of a grid connected photovoltaic (PV) and biomass Integrated renewable energy (IRE) system providing electricity to rural areas in the Beni Suef governorate, Egypt. The system load of the village is analyzed through the environmental and economic aspects. The model has been designed to provide an optimal system configuration based on daily data for energy availability and demands. A case study area, Monshaet Taher village (29° 1′ 17.0718”N, 30° 52′ 17.04”E) is identified for economic feasibility in this paper. HOMER optimization model plan imputed from total daily load demand, 2,340 kWh/day for current energy consuming of 223 households with Annual Average Insolation Incident on a Horizontal Surface of 5.79 (kWh/m^2/day) and average biomass supplying 25 tons / day. It is found that a grid connected PV-biomass IRE system is an effective way of emissions reduction and it does not increase the investment of the energy system.

Keywords: rural electrification, PV-biomass integrated optimal system, HOMER, feasibility study

1 Introduction

More than two-thirds of the populations of developing countries live in rural areas. There is a lack of fossil fuels in developing countries for the rural electrification and the funds for the development of these areas are limited. Hence, importing the needed resources will make the situation financially very untenable. Installation of modern energy systems improves access to potable water through pumping and distribution system. Enabling cold storage of medication and access to modern health care technologies can decrease the incidences of diseases. This in turn leads to reduced rates of child and maternal mortalities. It aids education and welfare of rural regions by providing adequate lighting and communication and relieving women of fuel and water collecting tasks and significantly contributes to improving gender equity. With modern renewable energy systems, it will be feasible to achieve ubiquitous access of electricity in the near future [1].

It is a known fact that rural population heavily depends on agriculture, and then uses traditional biomass resources extensively. Renewable energy sources such as solar, wind and water are abundantly available. Another essential comparative advantage of rural areas is open spaces that can be easily utilized to set up renewable energy systems. Hence, if possible to combine all these resources, in an efficient manner to fulfill all their needs, then the problem of rural development can be effectively confronted.

Since the place of study is a rural area where there is an abundant amount of solar energy and a lot of agricultural and animal waste, which is considered one of the most important sources of biomass energy, so the solar – biomass Integrated renewable energy system was chosen to provide electricity to the village.

There are a number of existing sizing methods used in the feasibility study of different hybrid systems based on various renewable energy resources (PV, wind and biomass), storage components (battery) and converters to make up of a hybrid system [2]. Compare the operational characteristics and cost values of three systems; stand-alone PV-wind-diesel, stand-alone PV-wind-hydrogen and grid connected PV-wind-hydrogen energy system. The comparisons prove that grid connected PV-wind-hydrogen energy system had the lowest total net present cost and cost of energy that makes it the most cost effective systems, and followed by PV-wind-diesel and stand-alone PV-wind-hydrogen system.

Reference [3] Examine the techno-economic aspects of replacing diesel generators and batteries of the system by hydrogen system as well as present the sizing optimization and simulation results of both systems. The results

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of the analyses also showed that the replacement of fossil fuel generator set with hydrogen technologies is technically feasible, but still not economically viable until reductions in the cost of hydrogen technologies are made in the future. In [4] a pre-feasibility study has been undertaken for the electrification of the rural off-grid village of Hurhudedanda, Nepal. The results of the analyses showed that PV is the most promising technology for Hurhudedanda. Three types of configurations were tested for the electrification of the entire village: Decentralized SHS, Centralized PV generation and Centralized diesel fueled Gen-set.

Recently, computer-based system simulation tools are used with increasing popular such as MATLAB, PSIM and HOMER. The HOMER (Hybrid Optimization Model for Electric Renewable) software package is widely used in determining probable configurations and implementation.

Analysis and modeling carried out proposing an optimal solution of a hybrid system of renewable energy by using the HOMER software for remote areas [5–12]. The Hybrid systems reported in these papers involve a combination of different energy sources like wind/battery, PV/battery, wind/PV/battery, wind/PV/diesel/battery. While [13] proposed a hybrid power system consisting of PV panels, a wind turbine and a biogas engine to supply the electricity demand of a village in Kenya. The results show that the hybrid system integrated with the biogas engine as backup can be a better solution than using a diesel engine as backup.

2 Study area and energy resources

In this research, Monshaet Taher village in the Beni Suef governorate has been chosen for the development of the integrated renewable energy system in Egypt. Beni Suef is an important agricultural trade center on the west bank of the Nile with total space estimated of 7,169 km.

Actually, the grid is available in Monshaet Taher nevertheless the supply of power is not reliable for supplying electric power continuously. Taking the advantages of abundant renewable energy resources that available, it will make the electricity more reliable. The total area of the village is about 1,463 acres, cultivated area represents, according to a statement of the Ministry of Agriculture 1,447 acres, representing 98.6 % of the area of the village. The village is characterized by the cultivation of crops, tomatoes, onions, strawberries and wheat.

In the Monshaet Taher village, there are about 223 households with 5,624 by local people [14]. Farming is the dominant source of income for rural households in this region. The remaining primary income source is the shop owner, petty trader and casual labor. The objective of this paper is to design an optimal economic power renewable energy system that feeds the required electric load of the Monshaet Taher village. The models have been designed to provide an optimal system configuration based on daily data for energy availability and demands.

A block diagram of the proposed integrated energy system is shown in Figure 1.

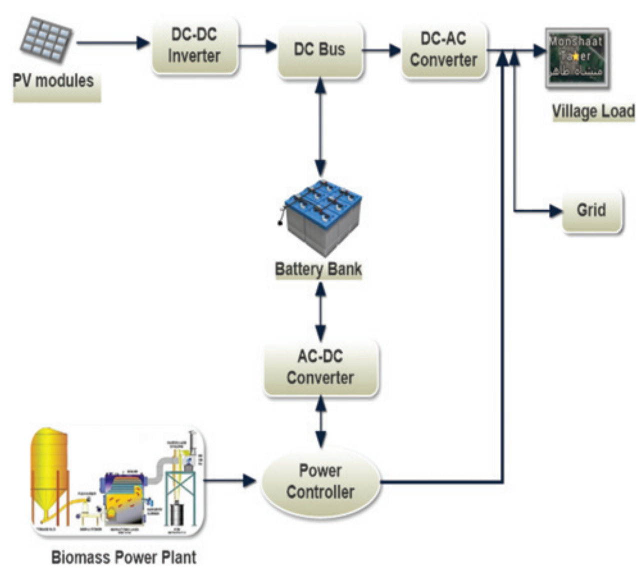

Figure 1: Proposed PV-biomass integrated system for the study area.

2.1 Load profiles of Monshaet Taher

The village consists of 223 houses and others public affairs. The electrical loads of the area are classified as domestic, agricultural and community. The domestic sector needs electricity for electrical appliances such as TV, fan and compact fluorescent lamps. The agricultural load includes water and irrigation pump. The community load includes schools, medical center and five mosques. The average load of the village is approximately 2,349 kWh/day (or 97.9 kW) with 172 kW peak and has a load factor of 0.570 (which equates to the average load divided by the peak load, the load factor is 97.9 kW/172 kW = 0.570).

2.2 Hourly load demand curves

Load demand curves have been performed by using a logical assumption referring that load demand varies in time, depends on the season, and depends on the inhabitant presence in a room; that is why load demand curves are erratic and quite choppy over the time. Figure 2 presents the typical profile of the electrical load which is used in the case study.

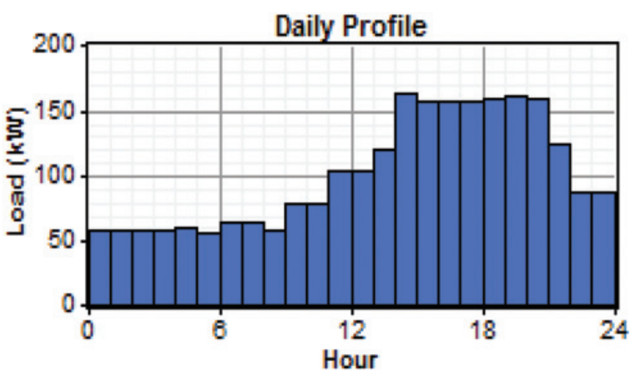

Figure 2: Daily load of the study area.

3 Renewable energy resources

3.1 Solar radiation

The solar resource used for the Monshaet Taher village at a location of 29°2′ N latitude and 31°6′ E longitude was taken from the NASA Surface Meteorology and Solar Energy website [15]. The annual average solar radiation was scaled to be 5.93 kWh/m²/Day. The scaled data of global radiation and a Clearness index of the study area are shown in Figure 3.

3.2 Biomass

Biomass can be defined as, the non-fossil organic materials that have an intrinsic organic chemical energy content, these materials, both liquid and solid, include vegetation and trees, municipal solid waste, municipal sewage, animal waste, and certain types of industrial waste [16].

Waste biomass in Egypt is generated from the following sources:

- Municipal solid waste;
- Agricultural residues (crop residues);
- Agro-industrial by-products (e. g. rice husk, bagasse);

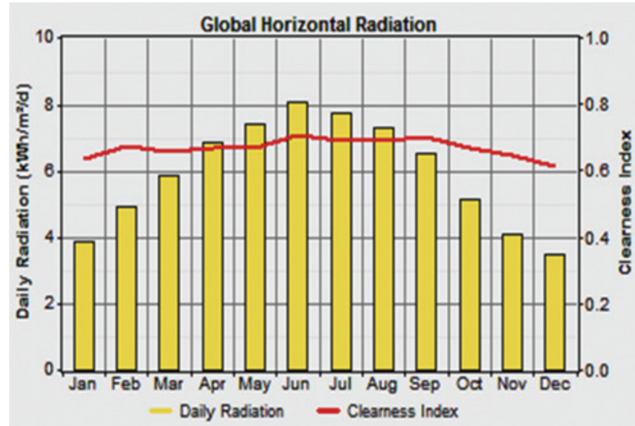

Figure 3: Monthly average global horizontal radiation and clearness index.

- Animal and poultry by-products (dung);
- Sewage sludge;
- Exotic plants (water hyacinth, reeds, etc.).

The total amount of municipal solid waste generated from Egypt's governorates in 2012 is estimated as 21 million tons/year [17]. The collection coverage of the municipal solid waste in rural areas ranges from 0 % to 30 % and in urban areas 50 % to 65 % [17]. For the final destination, 7 % of the municipal solid waste is currently composted, 10–15 % are recycled, 7 % landfilled and 80–88 % are open dumped.

Wastes produced due to agriculture activities are classified as [18]:

- Crop residues
- Pruning residues from trees and date palms
- Weeds and water weeds harvested from rivers, canals, and drains
- Food processing wastes
- Animal droppings.

Residue from agricultural production make up 30–60 % of the product that is used for human consumption and animal feed. An additional 30 % is made up of wastes from humans and animals.

3.2.1 Biomass gasification

3.2.1.1 Theory of gasification

The production of generator gas called gasification, is partial combustion of solid fuel (biomass) and takes place at temperatures of about 1,000 °C. The reactor is called a gasifier.

The combustion products from the complete combustion of biomass generally contain nitrogen, water vapor, carbon dioxide and surplus of oxygen. However, in gasification, where there is a surplus of solid fuel (incomplete combustion) the products of combustion are shown in Figure 4. The production of these gases is by reaction of water vapor and carbon dioxide through a glowing layer of charcoal. Thus the key to gasifier design is to create conditions such that a) biomass is reduced to charcoal and, b) charcoal is converted at a suitable temperature to produce CO and H_2 [19].

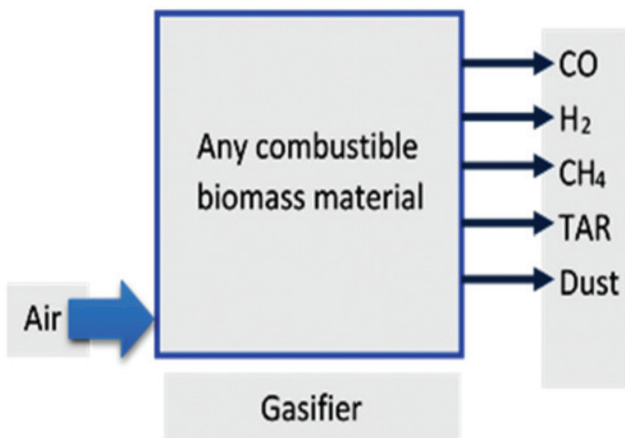

Figure 4: Products of gasification.

There are three types of gasifiers; Downdraft, Updraft and Crossdraft. The choice of one type of gasifier over another is dictated by the fuel, its final available form, its size, moisture content and ash content.

3.2.1.2 The suggested gasifier

Out of the different fixed bed gasifier configurations, the downdraught gasifier which is shown in Figure 5 have the potential for the best gas composition owing to their flow configuration, there is a better contact between the char bed and the products of oxidation, ensuring that a larger fraction of CO2 and H2O get reduced to CO, H2 and CH4, thus improving the calorific value of the gas [20].

3.2.2 Biomass material

The study focused on the use of agricultural crop residues, specifically maize residues. Maize residues (cobs, leaves and stalks) are abundantly available renewable materials that can be used as an energy source in

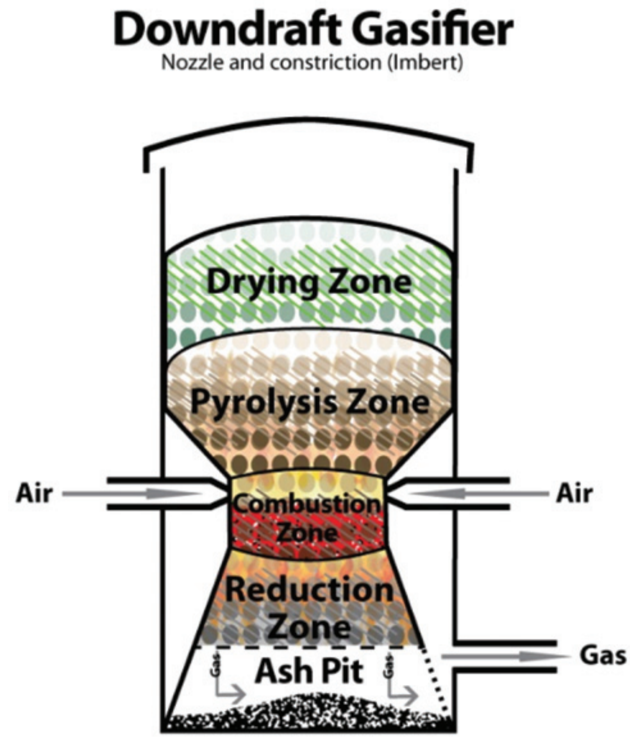

Figure 5: Downdraft gasifier [21].

gasification and combustion systems. Cobs, leaves and stalks are important residues of maize processing and consumption. For every 1 kg of dry maize grains produced, about 0.15 kg of cobs, 0.22 kg of leaves and 0.50 kg of stalks are produced [22]. Maize generates a substantial amount of residues, maize residues consist of the non-edible parts of the plant that are left on the farm after harvesting the targeted crop such as stalks, leaves and cobs [23]. Crop to Residue Ratio, Moisture Content and also, Lower Heating Value (LHV), and moisture content (%) of the maize residues is presented in Table 1.

Table 1: Crop to residue ratio, moisture content and LHV of maize crop [23].

Residue type	Crop to residue ratio	Moisture content (%)	LHV (MJ/kg)
Maize stalk	1	15.5	15
Maize cob	0.25	8	15

3.2.3 Biomass power plant

Biomass resource availability:

In Monshaet Taher the average data of available biomass supplying is about 25 tons/day as shown in Figure 6.

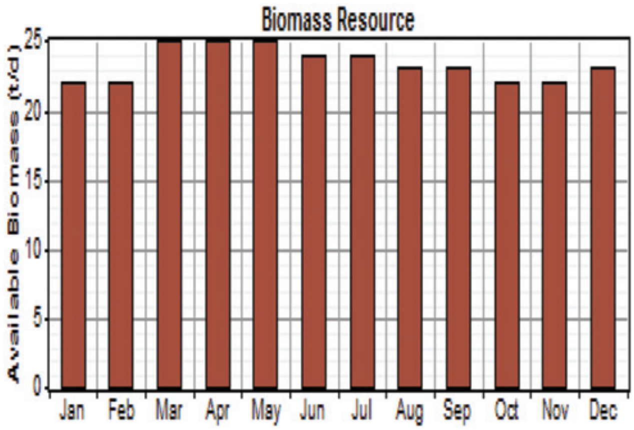

Figure 6: Biomass resources.

4 Research methodology

The feasibility studies through the modeling of economic and environmental characters of a PV-biomass integrated system with storages and converters. For economic aspect, the Net Present Cost (NPC) and Cost of electricity (COE) of the system are discussed and for the environmental aspect, the emission reduction and renewable fraction is investigated.

4.1 NPC and COE

HOMER uses NPC to represent the system's life cycle cost [24]. The NPC is calculated by equation 1:

$$NPC(\$) = \frac{TAC}{CRF} \tag{1}$$

Where TAC, the total annualized cost; CRF, the capital recovery factor, given by equation 2:

$$CRF(\$) = \frac{i(1+i)^N}{(1+i)^N - 1} \tag{2}$$

Where N, the number of years; i, the annual real interest rate (%).

The annual real interest rate is calculated by equation 3:

$$i = \frac{i' - f}{1+f} \tag{3}$$

where i', = the nominal interest rate; f, annual inflation rate.

The nominal interest rate and the annual inflation rate in Egypt as of 2016 are 11.75 % and 9.51 % respectively [25]. Therefore, the annual real interest rate is 2.0455 %.

COE is the average cost of useful electricity produced by the system in \$/kWh and can be calculated from equation 4 [26]:

$$COE(\$) = \frac{C_{ann,tot}}{E} \tag{4}$$

Where $C_{ann,tot}$ is the annual total cost, \$ and E is the total electricity consumption in *kWh /year*.

4.2 Renewable fraction (RF)

In a hybrid PV-biomass energy system, the renewable fraction specifies the contribution from different sources. PV fraction f_{PV} and biomass fraction f_{BIO} are given by eqs (5) and (6):

$$f_{PV} = \frac{E_{PV}}{E_{ann,tot}} \tag{5}$$

$$f_{BIO} = \frac{E_{BIO}}{E_{ann,tot}} \tag{6}$$

Where $E_{ann,tot}$, the annual total energy generation of the system. E_{PV} *and* E_{BIO} are respectively the PV and biomass generator electricity generation.

Then, knowing,

$$E_{PV} + E_{BIO} = E_{ann,tot} \tag{7}$$

4.3 Emissions

The main element of biogas is methane, therefore, the combustion of biogas must produce carbon dioxide (CO_2), given by equation 8:

$$E_m = Q_{BIO} \times F_{EM} \tag{8}$$

Where E_m is the emissions in kg/year, Q_{BIO} is the biomass consumption in kg and F_{EM} is the GHG emissions.

All calculations of those parameters and equations are carried out by HOMER software package.

5 Hybrid system modeling

The optimal system combination will lead to the optimal system design with the lowest levelized cost of energy. HOMER can optimize the system configuration, and perform sensitivity analyses, and therefore, the designer can make the right decision when the supplying load is needed to optimize with minimization of energy cost [27].

The input data, including solar and biomass resource data, electricity usage of the Monshaet Taher village and the components of a PV-Biomass system are put in the HOMER software tool built as a model plan shown in the following.

5.1 Optimization process

The goal of this paper is to minimize the NPC, the operating cost and the GHG emissions of the proposed system.

HOMER performs multiple optimizations under a range of input assumptions to gauge the effects of uncertainty or changes in the model inputs. Optimization determines the optimal value of the variables over which the system designer has control such as the mix of components that make up the system and the size or quantity of each [27].

A flowchart of the optimization process using HOMER is shown in Figure 7.

The IRE system consists of PV solar and biogas generator components. The system has also a component of current converter. Figure 8 shows a general scheme of the system in HOMER. The capital, replacement, operation and maintenance (O&M) costs and the search space of the various system component sizes have been given in Tables 2 and 3.

The cost of the components used in the proposed system was obtained as a result of a field survey of

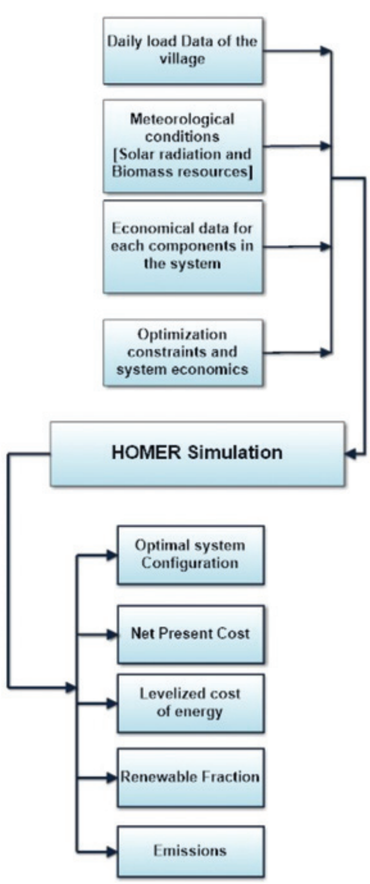

Figure 7: A flowchart of the optimization process using HOMER software.

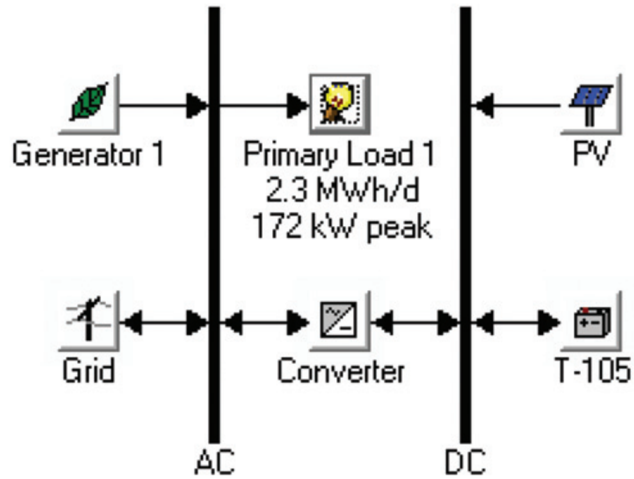

Figure 8: The optimum configuration model.

Table 2: Cost details of various equipment used in the proposed scheme.

		Capital ($)	Replacement ($)	O&M ($/yrs.)	Lifetime (yrs.)
PV module	1 kW	1,500	1,350	5	25
Battery: Trojan T-105	1 piece	145	145	10	Lifetime throughput 845 kWh
Biomass generator	50 kW	20,000	18,000	1 $/hr	15,000 h
Converter	3 kW	1,500	1,300	2	15

Table 3: The search space of the various components used in the proposed scheme.

PV array (kW)	Bio.Gen. (kW)	Grid (kW)	T-105 (Quantity)	Converter (kW)
0	0	0	10	0
50	50	100	20	100
100	100		30	200
150			40	250
200			50	
250				

local prices in Egypt and by searching companies producing component sites on the Internet.

5.2 Emissions/constraints

The emissions and constraints in running the system are described in Table 4 while the sensitivity inputs are presented in Table 5.

Table 4: Emissions and constraints of the system.

Carbon dioxide penalty	$ 0/t
Operating reserve as percentage of hourly load	0.1
Operating reserve as percentage of peak load	0
Operating reserve as percentage of solar power output	0.25
Operating reserve as percentage of wind power output	0.5

Table 5: Sensitivity inputs.

Sale (Cap. (kW))	Max. Cap. (Shortage (%))	Min. (RF (%))
50	0	0
100	5	50
	10	75

5.3 The grid information

The rate used is the current price of electricity in Egypt. The exact cost was used. The Egypt price for the residential sector was obtained from the electric utility [28] then converted to dollars and entered into HOMER. Currency conversion is assumed as 1$ = 10 Egyptian pounds.

- Grid rate price = 0.0305$/kWh
- sellback rate price = 0.0976 $/kWh.

Emission factors for grid power:

- Carbon dioxide = 632 g/kWh.
- Sulfur dioxide = 2.74 g/kWh.
- Nitrogen oxides = 1.34 g/kWh.

6 Simulation results and discussions

Simulations were run in the HOMER software for various configuration system types of the power system, in order

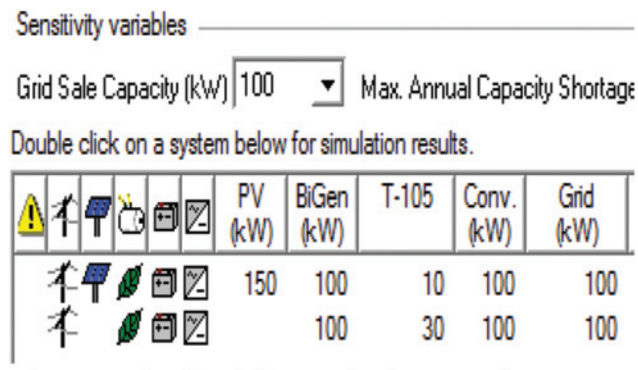

Figure 9: The simulation results for PV – biomass energy system.

to obtain the most efficient system configuration that would give the lowest net present cost to determine the basis of energy efficiency.

Figure 9 shows all the possible configurations that can be obtained using the various power sources.

Based on the results indicated in Figure 9, HOMER has been able to optimize the energy efficiency of the system using various conditions, by displaying the results from the most cost effective system for the least cost effective configuration. The summary of the optimal least cost hybrid system for the case study is described in Tables 6–8.

Table 6: System architecture of the optimized system.

PV array	150 kW
Biomass generator	100 kW
Grid	100 kW
Inverter	100 kW

Table 7: Cost summary of the optimized system.

Total net present cost	$ 462,992
Levelized cost of energy	$ 0.020/kWh
Operating cost	$ 7,546/year

Table 8: Electrical details of the optimized system.

Component	Production (kWh/year)	Fraction
PV array	284,482	23
Biomass generator	783,558	64
Grid purchases	156,657	13
Total	1,224,697	100

Table 9: The net present costs of the optimized system.

Component	Capital ($)	Replacement ($)	O&M($)	Salvage ($)	Total ($)
PV	225,000	0	14,565	0	239,565
Biomass generator	40,000	362,720	311,801	−13,454	701,067
Grid	0	0	−557,316	0	−557,316
Converter	1,450	2,151	1,942	−437	5,106
System	50,000	31,983	1,295	−8,707	74,571

Table 10: Annualized costs of the optimized system.

Component	Capital ($/year)	Replacement ($/year)	O&M ($/year)	Salvage ($/year)	Total ($/year)
PV	11,586	0	750	0	12,336
Biomass generator	2,060	18,678	16,056	−693	36,101
Grid	0	0	−28,699	0	−28,699
Converter	75	111	100	−23	263
System	2,575	1,647	67	−448	3,840

6.1 Details of optimized results

The summary of the optimal least cost integrated system for the case study is described in Tables 9 and 10.

It can be observed from Figure 10 and Table 8 that the Biomass generator is the dominant producer of electricity, for the selected system. The Biomass generator operates for 8,028 h (capacity factor 89.4 %), produces 783,558 kWh per year, with a total rated capacity of 100 kW. The PV panels produce 284,482 kWh/year, with a total rated capacity of 150 kW, operating for 4,383 hours/year (capacity factor of 21.7 %). The levelized cost of solar electricity is 0.0434 $/kWh.

The optimized system produced, 1,200,385 kWh/year, 71 % (857,386) of it fed to the AC primary load of the village while the other 29 % (342,999) of total electricity generated sales back to the grid. This demonstrates that this system has the capability in meeting also the demand growth in the future.

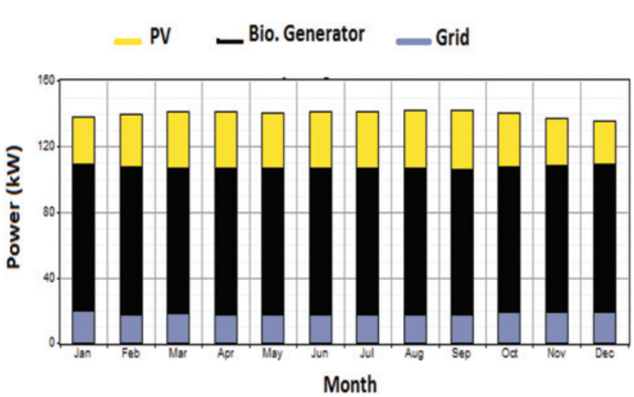

Figure 10: Monthly average electrical production.

Table 11 presents a Techno-Economic comparison between the two systems proposed by HOMER

Table 12 displays the Greenhouse Gases (GHG) emissions of the optimized system, the system sells more power to the grid than it buys from it over the year, the net grid purchases will be negative and so will the grid related emissions of each pollutant.

Table 11: Techno-economic comparison between the beat hybrids systems.

Configuration	Unit	Best optimize system	2nd optimize system
Solar PV	kW	150	0
Biomass generator	kW	100	100
Batteries (Trojan T-105)	no.	10	10
Biomass generator (Hours of operation)	hr./year	8,028	7,827
Converter	kW	100	100
Grid	kW	100	100
Emissions (Carbon dioxide)	kg/year	−117,636	47,333
Renewable fraction	%	86.9	77.2
Total capital cost	$	316,450	91,450
Total NPC	$	462,992	603,252
Levelized cost of energy	$/kWh	0.02	0.031
Operating cost	$ /year	7,546	26,355

Table 12: Greenhouse gases (GHG) emissions.

Quantity	Value (kg/yr.)
Carbon dioxide	−117,636
Carbon monoxide	4.96
Unburned hydrocarbons	0.55
Particulate matter	0.374
Sulfur dioxide	−511
Nitrogen oxides	−205

Table 13: Comparison between the proposed optimized system and grid connected system.

Item	The proposed optimized system	Grid connected system
Configuration	PV/ biomass generator	Grid only
Total capital cost ($)	316,450	$ 0
Total NPC ($)	462,992	$ 507,827
Levelized cost of energy ($/kWh)	0.02	0.031
Operating cost ($ /year)	7,546	26,150
Carbon dioxide Emissions (kg/ year)	−117,636	541,868

It can be observed from Table 13 that the proposed optimized system is better and more economical than the grid connected system. From an environmental point of view, in terms of Greenhouse Gases (GHG) Emissions, the proposed hybrid PV-biomass system is preferred over the grid connected system.

7 Conclusions

This paper analyzed the feasibility of grid-connected IRE system to supply a small village in Egypt, the results show the least-cost combination of solar PV, biomass generator and batteries that can meet the demand in a dependable manner at a cost of $ 0.02/kWh. Given the availability of solar power at this location, most of the electricity in the optimal solution comes from the Biomass generator and it provides a cheap source of power to the locality.

From the simulation results (costs and emissions), it is clear that the use of hybrid PV/biomass system achieves significantly lower NPC and reduction of CO2, as compared to a grid connected system. It is found that the renewable energy sources are very much feasible to solve the rural demand of Egypt.

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